Android Program

1.Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

Code part:

```
<?xml version="1.0" encoding="utf-8"?>


<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:orientation="vertical"

    android:padding="20dp"

    android:layout_width="match_parent"

    android:layout_height="match_parent">


    <EditText

        android:id="@+id/nameText"

        android:hint="Enter your name"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"/>


    <Button

        android:id="@+id/okButton"

        android:text="OK"

        android:layout_width="match_parent"

        android:layout_height="wrap_content"/>


    <TextView

        android:id="@+id/resultText"

        android:textSize="20sp"

        android:layout_marginTop="20dp"
```

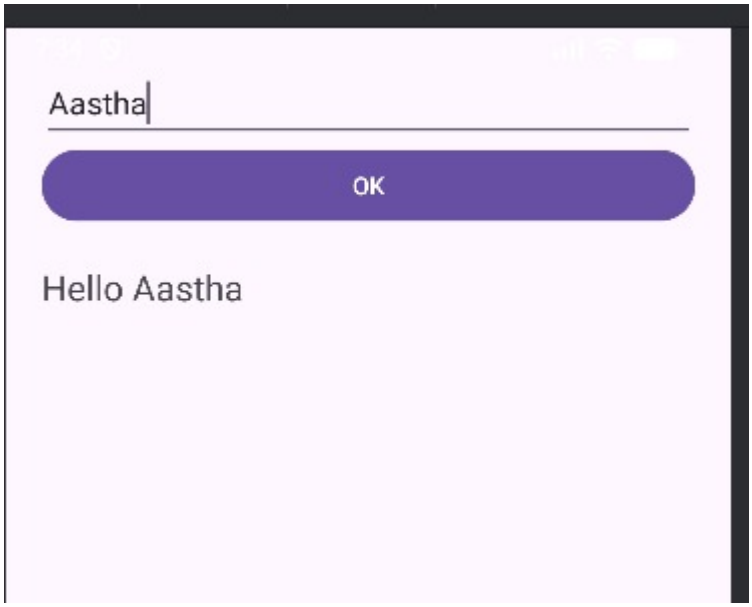
```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Java code:

```
package com.game.helloapp;



import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText nameText;
    Button okButton;
    TextView resultText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        nameText = findViewById(R.id.nameText);
        okButton = findViewById(R.id.okButton);
```

```
        resultText = findViewById(R.id.resultText);

        okButton.setOnClickListener(v -> {

            String name = nameText.getText().toString();

            resultText.setText("Hello " + name);

        });
    }
}
```

output:



2. Create an Android application that generates random arithmetic questions and allows the user to select True or False. Display the score based on correct answers.

Coding part:

```
ml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tvTime"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="TIME: 05"
        android:textColor="#FF0000"
        android:textSize="34sp"
        android:layout_margin="16dp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"/>

    <TextView
        android:id="@+id/tvScore"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="SCORE: 00"
        android:textColor="#FF0000"
        android:textSize="34sp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_margin="16dp"/>

    <TextView
        android:id="@+id/tvQuestion"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="11 + 64 = 75"
        android:textSize="36sp"
        app:layout_constraintTop_toBottomOf="@id/tvTime"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

    <Button
```

```
        android:id="@+id/btnTrue"
        android:layout_width="0dp"
        android:layout_height="60dp"
        android:text="✓"
        android:textSize="24sp"
        android:textColor="#FFFFFF"
        android:backgroundTint="#D32F2F"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toStartOf="@id/btnFalse"
        app:layout_constraintWidth_percent="0.5"/>

    <Button
        android:id="@+id/btnFalse"
        android:layout_width="0dp"
        android:layout_height="60dp"
        android:text="✕"
        android:textSize="24sp"
        android:textColor="#FFFFFF"
        android:backgroundTint="#D32F2F"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toEndOf="@id/btnTrue"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintWidth_percent="0.5"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

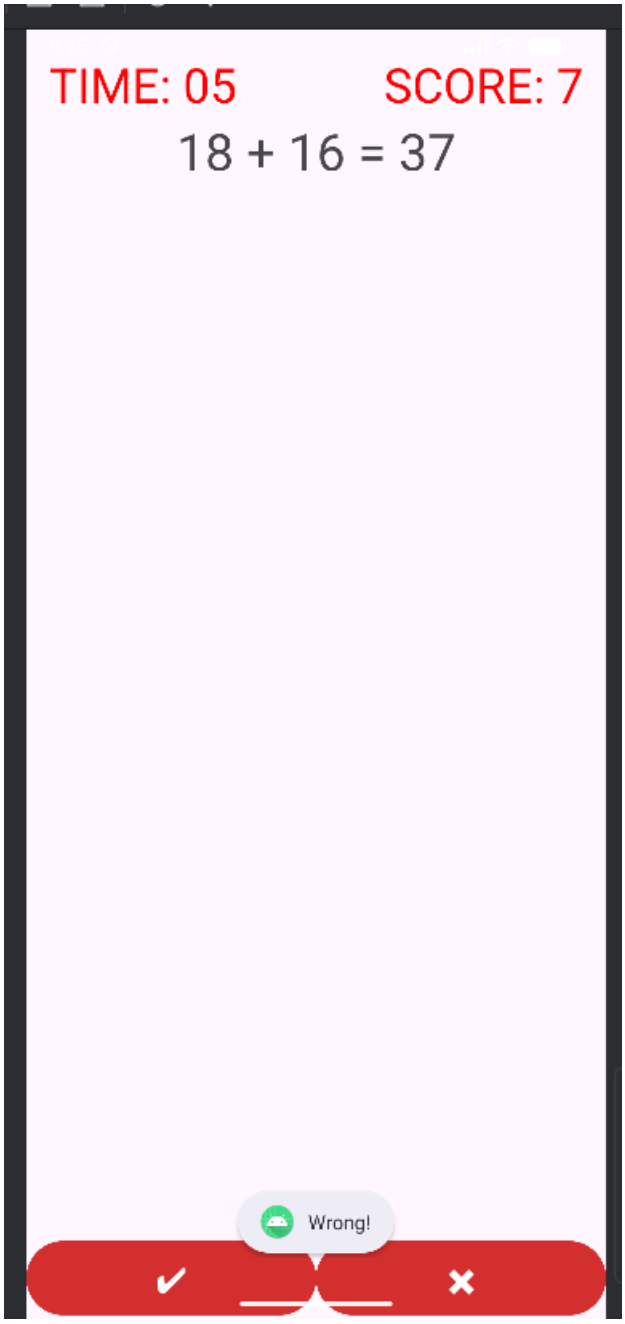
Java code:

```
package com.game.myapplication;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.Random;

public class MainActivity extends AppCompatActivity {
```

```
TextView tvQuestion, tvScore, tvTime;
Button btnTrue, btnFalse;

int score = 0;
int correctAnswer;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    tvQuestion = findViewById(R.id.tvQuestion);
    tvScore = findViewById(R.id.tvScore);
    tvTime = findViewById(R.id.tvTime);
    btnTrue = findViewById(R.id.btnTrue);
    btnFalse = findViewById(R.id.btnFalse);

    generateQuestion();

    btnTrue.setOnClickListener(view -> checkAnswer(true));
    btnFalse.setOnClickListener(view -> checkAnswer(false));
}

void generateQuestion() {
    Random random = new Random();

    int a = random.nextInt(50);
    int b = random.nextInt(50);

    correctAnswer = a + b;

    boolean isCorrect = random.nextBoolean();

    int shownAnswer = isCorrect ? correctAnswer : correctAnswer + random.nextInt(10);

    tvQuestion.setText(a + " + " + b + " = " + shownAnswer);
}

void checkAnswer(boolean userAnswer) {
    boolean actual =
```

```
tvQuestion.getText().toString().contains(String.valueOf(correctAnswer));

        if (userAnswer == actual) {
            score++;
            Toast.makeText(this, "Correct!", Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(this, "Wrong!", Toast.LENGTH_SHORT).show();
        }

        tvScore.setText("SCORE: " + score);
        generateQuestion();
    }
}
```

output:



3. Design an Android app that uses MediaPlayer to play an audio file stored in the raw folder.

Coding:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
```

```
android:padding="24dp">

<TextView
    android:id="@+id/titleTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Music Player"
    android:textSize="30sp"
    android:textStyle="bold"
    android:layout_marginBottom="30dp"/>

<Button
    android:id="@+id/btnPlay"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Play"
    android:onClick="playMusic"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/btnPause"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Pause"
    android:onClick="pauseMusic"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/btnStop"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Stop"
    android:onClick="stopMusic"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/btnRestart"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Restart"
    android:onClick="restartMusic"
```

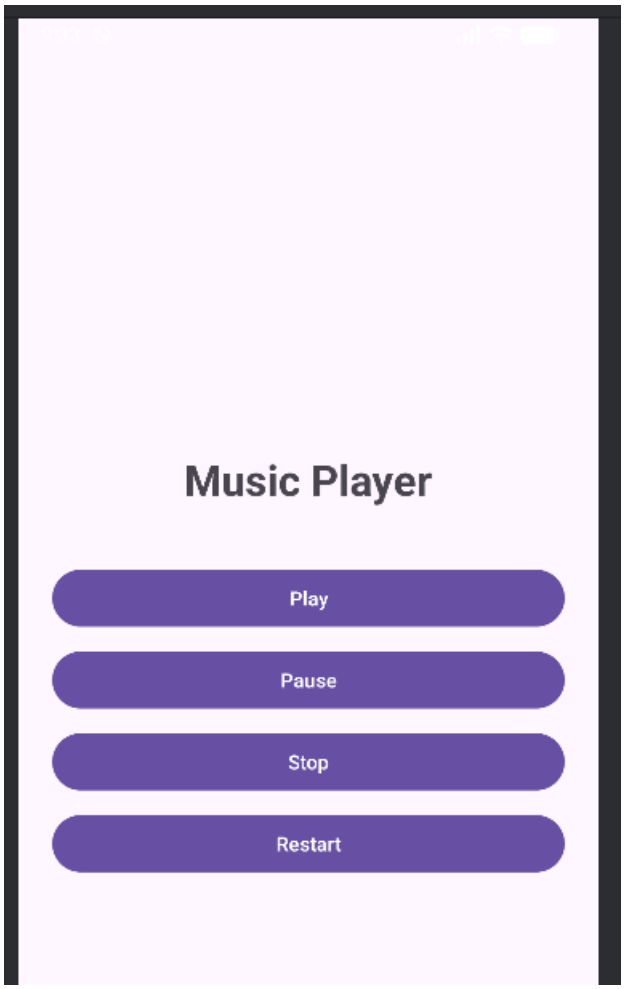
```
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

Java part:

```
package com.game.mymusic;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    MediaPlayer mediaPlayer;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        mediaPlayer = MediaPlayer.create(this, R.raw.song);
    }

    public void playMusic(View view) {
        if (mediaPlayer != null) {
            mediaPlayer.start();
            Toast.makeText(this, "Playing Music", Toast.LENGTH_SHORT).show();
        }
    }

    public void pauseMusic(View view) {
        if (mediaPlayer != null && mediaPlayer.isPlaying()) {
            mediaPlayer.pause();
            Toast.makeText(this, "Music Paused", Toast.LENGTH_SHORT).show();
        }
    }

    public void stopMusic(View view) {
```

```
        if (mediaPlayer != null) {
            mediaPlayer.stop();
            mediaPlayer = MediaPlayer.create(this, R.raw.song);
            Toast.makeText(this, "Music Stopped", Toast.LENGTH_SHORT).show();
        }
    }
}
```

Output:



4. Design an Android app to add and display a list of service providers (like Electrician, Plumber) using ListView.

Coding:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
```

```
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/serviceName"
        android:hint="Service Type (Electrician / Plumber)"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/personName"
        android:hint="Worker Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/serviceContact"
        android:hint="Contact Number"
        android:inputType="phone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/addService"
        android:text="Add Service"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <ListView
        android:id="@+id/serviceList"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

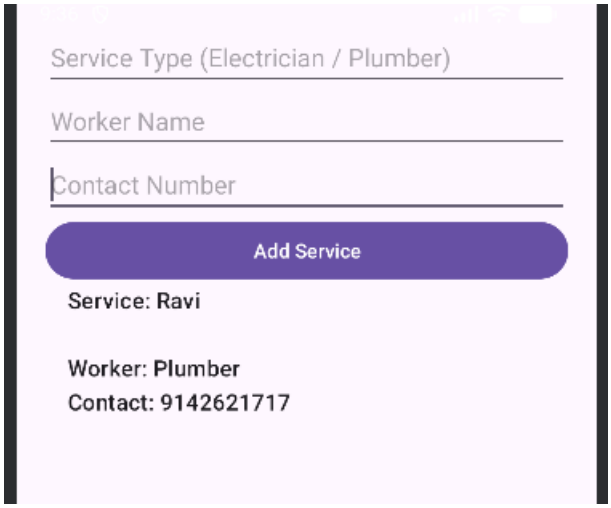
Java part:

```
package com.game.serviceapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
import android.widget.*;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    EditText serviceName, personName, serviceContact;
    Button addService;
    ListView serviceList;

    ArrayList<String> services;
    ArrayAdapter<String> adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        serviceName = findViewById(R.id.serviceName);
        personName = findViewById(R.id.personName);
        serviceContact = findViewById(R.id.serviceContact);
        addService = findViewById(R.id.addService);
        serviceList = findViewById(R.id.serviceList);

        services = new ArrayList<>();

        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                services
        );

        serviceList.setAdapter(adapter);

        addService.setOnClickListener(v -> {

            String service = serviceName.getText().toString();
            String name = personName.getText().toString();
            String contact = serviceContact.getText().toString();

            if(service.isEmpty() || name.isEmpty() || contact.isEmpty())
```

```
            {
                Toast.makeText(this,"Fill all fields",Toast.LENGTH_SHORT).show();
                return;
            }

            String data =
                    "Service: " + service +
                            "\nWorker: " + name +
                            "\nContact: " + contact;

            services.add(data);

            adapter.notifyDataSetChanged();

            serviceName.setText("");
            personName.setText("");
            serviceContact.setText("");

            Toast.makeText(this,"Service Added",Toast.LENGTH_SHORT).show();

        });

        serviceList.setOnItemLongClickListener((parent, view, position, id) -> {

            services.remove(position);
            adapter.notifyDataSetChanged();

            Toast.makeText(this,"Service Removed",Toast.LENGTH_SHORT).show();

            return true;

        });

    }
}
```

Output:



5.GridImage

Coding part:

```
<?xml version="1.0" encoding="utf-8"?>
<GridLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:rowCount="2"
    android:columnCount="2"
    android:padding="16dp">

    <!-- Red Button -->
    <Button
        android:id="@+id/btnRed"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_rowWeight="1"
        android:layout_columnWeight="1"
        android:text="RED"
        android:backgroundTint="@android:color/holo_red_dark"
        android:textColor="@android:color/white"/>

    <!-- Blue Button -->
    <Button
        android:id="@+id/btnBlue"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_rowWeight="1"
```

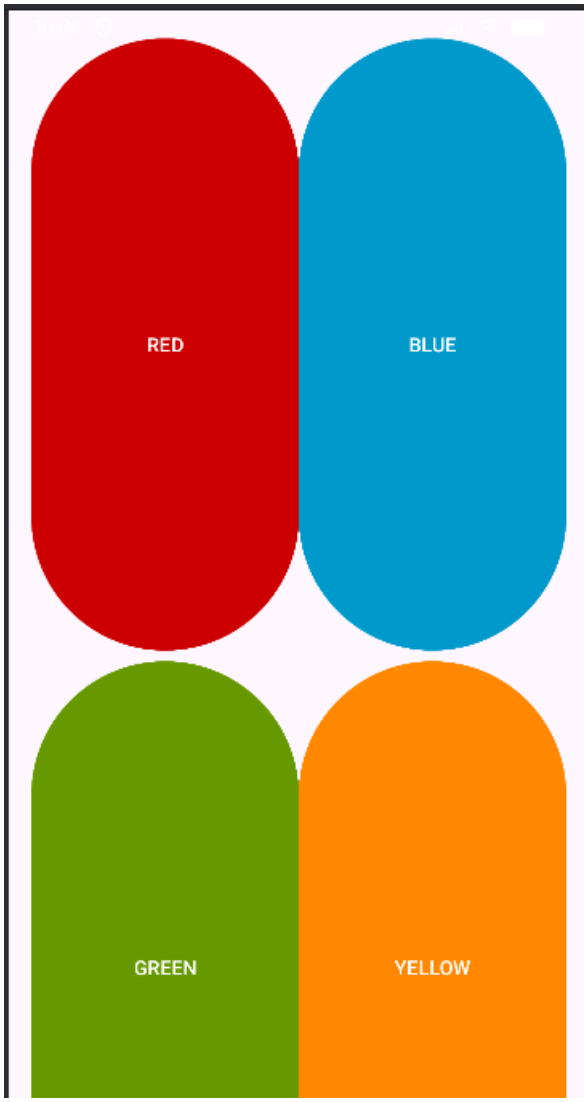
```
        android:layout_columnWeight="1"
        android:text="BLUE"
        android:backgroundTint="@android:color/holo_blue_dark"
        android:textColor="@android:color/white"/>

    <!-- Green Button -->
    <Button
        android:id="@+id/btnGreen"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_rowWeight="1"
        android:layout_columnWeight="1"
        android:text="GREEN"
        android:backgroundTint="@android:color/holo_green_dark"
        android:textColor="@android:color/white"/>

    <!-- Yellow Button -->
    <Button
        android:id="@+id/btnYellow"
        android:layout_width="0dp"
        android:layout_height="0dp"
        android:layout_rowWeight="1"
        android:layout_columnWeight="1"
        android:text="YELLOW"
        android:backgroundTint="@android:color/holo_orange_dark"
        android:textColor="@android:color/white"/>

</GridLayout>
```

Java part:

```
package com.game.gridimage;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import com.game.gridimage.R;
```

```
public class MainActivity extends AppCompatActivity {

    Button btnRed, btnBlue, btnGreen, btnYellow;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnRed = findViewById(R.id.btnRed);
        btnBlue = findViewById(R.id.btnBlue);
        btnGreen = findViewById(R.id.btnGreen);
        btnYellow = findViewById(R.id.btnYellow);

        btnRed.setOnClickListener(v ->
                Toast.makeText(MainActivity.this, "Red Button Clicked",
Toast.LENGTH_SHORT).show()
        );

        btnBlue.setOnClickListener(v ->
                Toast.makeText(MainActivity.this, "Blue Button Clicked",
Toast.LENGTH_SHORT).show()
        );

        btnGreen.setOnClickListener(v ->
                Toast.makeText(MainActivity.this, "Green Button Clicked",
Toast.LENGTH_SHORT).show()
        );

        btnYellow.setOnClickListener(v ->
                Toast.makeText(MainActivity.this, "Yellow Button Clicked",
Toast.LENGTH_SHORT).show()
        );
    }
}
```

output:


RED
BLUE
GREEN
YELLOW